Assignment 2: Build a Low-Poly Character Pipeline from Blender to Unity

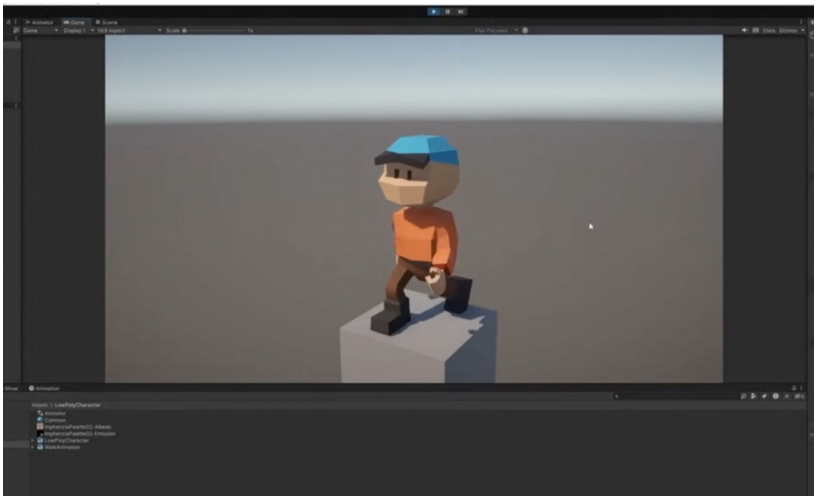
Overview

In this assignment, you will complete a full low-poly character pipeline that connects **Blender** and **Unity**. Based on the workflow introduced in **Lab 6**, you will model, rig, animate, and export a character in Blender, then import it into Unity and present the result clearly.

This assignment focuses on helping you understand how 3D assets move across software tools in a complete and practical workflow. By completing this process, you will strengthen your skills in character creation, animation, asset export, and Unity presentation.

After completing the required pipeline, you are encouraged to personalize your character by adding your own visual and creative ideas.

Reminder



For this lab, you only need to complete **one character, Bob.** You do not need to create the additional stylized versions shown in the video, such as the female character or warrior. Your submitted Blender file should include the completed character, rigging, and walk cycle animation. In Unity, your character should be able to play the **walk cycle animation** correctly.

Assignment Tasks

Part 1: Recreate the Core Pipeline (Tutorial Replication)

Follow the tutorial materials provided in **Lab 6** and reproduce the following pipeline:

- Model **one low-poly humanoid character, Bob**, in Blender
- Apply basic colorization using the provided palette texture
- Rig the character with a basic armature and leg IK
- Skin the character and adjust weight painting where needed
- Create a **walk cycle animation**
- Export the character and animation from Blender as an **.fbx** file
- Import the character into Unity correctly
- Create and assign the material/texture in Unity
- Configure the animation clip properly (e.g., loop settings if needed)
- Create a Unity build that clearly presents the imported character with the **walk cycle playing correctly**

Part 2: Add Your Creative Ideas

After completing the core pipeline, personalize your character by adding your own creative ideas. You may choose from the following suggestions, or develop your own original approach:

- Improve the Unity presentation (e.g., better lighting, scene setup, or designing a simple platform/environment for the character to walk on)
- Modify the character's appearance (e.g., colors, body proportions, clothing details)

- Add simple accessories or props (e.g., glasses, hat, bag, tool)
- Add small visual details to make the character more distinctive

Your creativity will be evaluated based on **clarity, coherence, and originality**, not on complexity alone.

Evaluation

The total points of Assignment 2 are up to 10 points.
Total score = basic points (7 pts) + bonus points (3 pts).

Basic

1. Model one low-poly character successfully in Blender, with correct basic colorization/material setup — 1 point
2. Rig the character correctly and complete skinning/weight painting so that the body deforms properly during animation — 1 point
3. Create a walk cycle animation that is complete, clear, and plays correctly — 2 points
4. Export the character correctly from Blender and import it into Unity successfully, with the character and materials shown properly — 2 points
5. Present the final result clearly in Unity, with the walk cycle animation playing correctly in the executable build — 1 point

Bonus

After completing all basic requirements, you are encouraged to further develop your project in any creative direction you find meaningful. You are free to introduce additional elements, visual ideas, animation variations, environmental design, interaction logic, or any other extensions that enhance the overall experience. There is no restriction on the type of innovation, as long as it meaningfully improves or enriches the final result. Each clearly implemented and effective innovation may earn 1 bonus point, up to a maximum of 3 bonus points. All bonus features must be clearly visible in the executable build and briefly described in the README.txt. Improvements that are not visible or not documented will not be counted toward bonus points.

Important Notes on Evaluation

Please note that grading will be based primarily on the **executable build (.exe / .app)** and the README file. The Unity project file and Blender source file will be used for reference when needed and may not always be opened during grading.

Therefore, it is your responsibility to ensure that the executable build functions correctly and clearly demonstrates your final result. The build should show the character properly in Unity and present the walk cycle animation clearly for evaluation. Any features or improvements not shown in the executable build may not be included in the score.

If you make any improvements for bonus points, please make sure they are both:

- clearly visible in the executable build, and
- briefly described in the README.txt.

Submission Requirements

You are required to submit the following 4 items to the Assignment 2 portal on Blackboard:

- Blender file (.blend)
- Unity project folder (.zip)
- Unity executable build folder (.zip)
- README document (.txt)

Please do not combine these files into a single compressed package. Each file should be uploaded individually.

1. Blender file (.blend)

   - Submit the original Blender file used to model, rig, and animate your character.
   - File name format: YourName&ID_Assignment2.blend

2. Unity project folder (.zip)

   - Compress your full Unity project folder into a .zip file.
   - This should include all assets, scenes, imported FBX files, materials, textures, animation settings, and any other related files used in your project.
   - File name format: YourName&ID_UnityProject.zip

3. Unity executable build folder (.zip)

After building your project in Unity, you will get a folder that contains your executable build file.

- Compress this entire folder into a .zip file.
- File name format: YourName&ID_Build.zip
- This is the version we will open and test during grading.

4. README document (.txt)

Include a plain text file named README.txt that clearly states:

- the operating system used to develop and run the project (Windows or macOS)
- if you are using macOS, please clearly indicate that the project should be opened as a macOS build
- if you made any improvements for bonus points, briefly describe them in your `README.txt`

Due Date

- **23:59, Mar. 13th (Friday)**
- Late submissions will result in a point deduction: 1 point per day of delay (e.g., 1 point for one day late, 2 points for two days late, and so on).